C++

程式語言（二）

Introduction to Programming (II)

Dynamic Memory Allocation

Joseph Chuang-Chieh Lin

Dept. CSE, NTOU

Platform/IDE

- Dev-C++

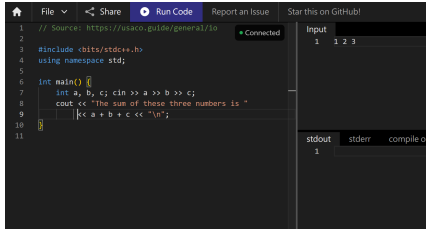
https://www.pngegg.com/en/search?q=Dev-C

- Codeblocks

https://icons8.com/icons/set/code-blocks

- OnlineGDB (https://www.onlinegdb.com/)

- Real-Time Collaborative Online IDE (https://ide.usaco.guide/)



Textbooks (We focusing on C++11)

- ***Learn C++ Programming by Refactoring* (由重構學習 C++ 程式設計). Pang-Feng Liu (劉邦鋒). NTU Press. 2023.**
- ***C++ Primer. 5th Edition.* Stanley B. Lippman, Josée Lajoie, Barbara E. Moo. 2019.**
- *Effective C++*. Scott Meyers. O'Reilly. 2016.
- *Thinking in C++. Vol. 1: Introducing to Standard C++*. 2nd Edition. Bruce Eckel. Prentice Hall PTR. 2000.

Useful Resources

- Tutorialspoint
  - https://www.tutorialspoint.com/cplusplus/index.htm
  - Online C++ Compiler
- Programiz
  - https://www.programiz.com/cpp-programming
- LEARN C++
  - https://www.learncpp.com/
- MIT OpenCourseWare - Introduction to C++
  - https://ocw.mit.edu/courses/6-096-introduction-to-c-january-iap-2011/pages/lecture-notes/
- Learning C++ Programming
  - https://www.programiz.com/cpp-programming
- GeeksforGeeks
  - https://www.geeksforgeeks.org/c-plus-plus/

Dynamic Memory Allocation in C++

Purpose of using dynamic memory

- Properly freeing dynamic objects turns out to be a surprisingly rich source of bugs.
- Programs tend to use dynamic memory for one of three purposes:
  1. They don’t know how many objects they’ll need.
  2. They don’t know the precise type of the objects they need.
  3. They want to share data between several objects.

Purpose of using dynamic memory

- Properly freeing dynamic objects turns out to be a surprisingly rich source of bugs.

- Programs tend to use dynamic memory for one of three purposes:
  1. They don’t know how many objects they’ll need.
  2. They don’t know the precise type of the objects they need.
  3. They want to **share data between several objects**.

Sharing data between objects (Example)

```
class Board {
public:
    int score = 0;
};

class Player {
public:
    Board* board; // points to a shared Board
    Player(Board* b) : board(b) {}

    void addScore() {
        board->score++;
    }
};
```

Usage:

```
Board* b = new Board();

Player p1(b);
Player p2(b);

p1.addScore();
cout << b->score << endl; // 1
p2.addScore();
cout << b->score << endl; // 2
```

https://onlinegdb.com/mIw1mqHhL

- `p1`, `p2`: two different `Player` objects.
  - Both point to the same `Board`. So the data is **shared**.

Note & Remark

- Sharing data can also be done using "`static`".

```
class Player {
public:
    static int totalScore;
};

int Player::totalScore = 0;
```

```
Player a, b;
a.totalScore = 10;
cout << b.totalScore << endl; // 10
```

Note & Remark

- Sharing data can also be done using "`static`".

```
class Player {
public:
    static int totalScore;
};

int Player::totalScore = 0;
```

```
Player a, b;
a.totalScore = 10;
cout << b.totalScore << endl; // 10
```

- However,
  - A static object usually has a lifetime for the entire program.
  - It is fixed and exists independently of how many objects you create.

Note & Remark

- Sharing data can also be done using "`static`".

```
class Player {
public:
    static int totalScore;
};

int Player::totalScore = 0;
```

```
Player a, b;
a.totalScore = 10;
cout << b.totalScore << endl; // 10
```

- However,
  - Dynamic memory lets you create shared objects **when needed**, in any number, and **control their lifetime more flexibly**.

`new` and `delete`?

- In C++, people are used to use `new` operator (cf., `malloc()` in C) to allocate memory and `delete` (cf., `free()` in C) to free memory allocated by `new`.
- However, using these operators to manage memory is considerably more error-prone.
- From C++11 and newer versions, we are encouraged to use **smart pointers** to manage dynamic objects.
  - They are safer and easier.

Smart Pointers (the `shared_ptr` class)

```
shared_ptr<string> p1;
unique_ptr<int> p2;
```

*Actually there is also `make_unique` but it's in C++14 standard.

```
//use make_shared function
shared_ptr<int> p3 = make_shared<int>(42);
//42
shared_ptr<string> p4 = make_shared<string>(10, '9');
//9999999999
shared_ptr<int> p5 = make_shared<int>();
```

Smart Pointers (the `shared_ptr` class)

```
shared_ptr<string> p1;
unique_ptr<int> p2;
```

*Actually there is also `make_unique` but it's in C++14 standard.

```
//use make_shared function
shared_ptr<int> p3 = make_shared<int>(42);
//42
shared_ptr<string> p4 = make_shared<string>(10, '9');
//9999999999
shared_ptr<int> p5 = make_shared<int>();
```

```
//we can also use "auto"
auto p3 = make_shared<int>(42);
//42
auto p4 = make_shared<string>(10, '9');
//9999999999
auto p5 = make_shared<int>();
```

An Example

https://onlinegdb.com/dSS35GJ2l

```
#include <iostream>
#include <memory>

using namespace std;

class Grade {
private:
    int math;
    int eng;
    int sum;
public:
    Grade() = default;
    Grade(int m, int e): math(m), eng(e) {};
    ~Grade() { cout << "destructor of 'Grade' works here" << endl; } ;
    void SumUp() { sum = math + eng; }
    int ShowSum() { return sum; }
};
```

```
int main()
{
    auto ptr = make_shared<Grade>(100, 90);
    ptr->SumUp();
    cout << "The total grades: "
         << ptr->ShowSum() << endl;
    return 0;
}
```

```
The total grades: 190
destructor of 'Grade' works here
```

Copying and Assigning `shared_ptr`

https://onlinegdb.com/i2OgvL1kL

- When we copy or assign a `shared_ptr`, each `shared_ptr` keeps track of **how many** other `shared_ptrs` point to the same object.

```
auto p = make_shared<int>(42); // object to which p points has one user
auto q(p); // p and q point to the same object; q is a copy of p
// object to which p and q point has two users

auto r = make_shared<int>(42); // new object; r has its own separate
                               // shared_ptr with one user
r = q; // r is assigned to q, so now r points to the same object as q
// The object r previously pointed to has no users and is freed
// The object now pointed to by p, q, and r has 3 users

cout << r.unique(); // 0; print out whether p.use_count() is 1 or not
cout << r.use_count(); // 3; print out number of objects sharing with r
```

More on shared_ptr

- shared_ptr's **automatically**
  - **destroy** their objects (by a destructor of the class).
  - **free** the associated memory.

```
// factory returns a shared_ptr pointing to a dynamically allocated object
shared_ptr<Foo> factory(T arg) {
    // process arg as appropriate
    // shared_ptr will take care of deleting this memory
    return make_shared<Foo>(arg);
} // the object will be appropriately deleted with the allocated memory freed
```

```
void use_factory(T arg) {
    shared_ptr<Foo> p = factory(arg); // use p
} // p goes out of scope; the memory to which p points is automatically freed
```

More on shared_ptr

- shared_ptr's **automatically**
  - **destroy** their objects (by a destructor of the class).
  - **free** the associated memory.

```
// factory returns a shared_ptr pointing to a dynamically allocated object
shared_ptr<Foo> factory(T arg) {
    // process arg as appropriate
    // shared_ptr will take care of deleting this memory
    return make_shared<Foo>(arg);
}
```

```
// factory returns a shared_ptr pointing to a dynamically allocated object
shared_ptr<Foo> use_factory(T arg) {
    shared_ptr<Foo> p = factory(arg); // use p
    return p; // reference count is incremented when we return p
} // p goes out of scope; the memory to which p points is NOT freed
```

Managing Memory Directly (new & delete)

```
int *pi = new int;
string *ps = new string;
int *pi = new int(1024);
string *ps2 = new string(10, '9');
// allocate and initialize a const int
const int *pci = new const int(1024);
// allocate and initialize an empty string
const string *pcs = new const string;
```

```
int i, *pi1 = &i, *pi2 = nullptr;
double *pd = new double(33), *pd2 = pd;
delete i; // error: i is not a pointer
delete pi1; // undefined: pi1 refers to a local
delete pd; // ok
delete pd2; // undefined: the memory pointed to by pd2 was already freed
delete pi2; // ok: it is always ok to delete a null pointer
```

Using shared_ptrs with new

```
shared_ptr<double> p1;
shared_ptr<int> p2(new int(42)); //direct initialization
```

Note that the following initialization is wrong:

```
shared_ptr<int> p1 = new int(42);
//error: this is copy initialization, we must use direct initialization
```

Note that the following implicit conversion is also wrong:

```
shared_ptr<int> clone(int p) {
    return new int(p);
}
```

correction

```
shared_ptr<int> clone(int p) {
    return shared_ptr<int>(new int(p));
}
```

Dynamic Arrays

```
int *pia = new int[10]; // uninitialized 10 ints
int *pia2 = new int[10](); //initialized to be 10 0's;
string *psa = new string[10]; // block of 10 empty strings
string *psa2 = new string[10]() ; // block of 10 empty strings
int *pia3 = new int[5]{0,1,2,3,4};
string *psa3 = new string[10]{"a", "b", string(3,'x')};
// the first three elements are initialized from given initializers
// remaining elements are value initialized
```

```
// Freeing dynamic arrays
delete [] pia;
delete [] psa;
...
```

*Remark

- Using a library container (e.g., vector, see STL in the future lectures, if it's possible) is better (safer, easier, and more efficient) and even more pronounced under the new standard.

```
vector<int> v1(10); // v1 has 10 elements with value 0
vector<int> v2(10, 1); // v2 has 10 elements with value 1
vector<int> v3{1, 2, 3}; // v3 has two elements with values 1, 2, and 3
v1.push_back(9); // add 9 into the rear of v1
...
```

Exercise:
Try to use `new` and `delete` instead

https://onlinegdb.com/dSS35GJ2l

```
#include <iostream>
#include <memory>

using namespace std;

class Grade {
private:
    int math;
    int eng;
    int sum;
public:
    Grade() = default;
    Grade(int m, int e): math(m), eng(e) {};
    ~Grade() { cout << "destructor of 'Grade' works here" << endl; } ;
    void SumUp() { sum = math + eng; }
    int ShowSum() { return sum; }
};
```

```
int main()
{
    auto ptr = make_shared<Grade>(100, 90);
    ptr->SumUp();
    cout << "The total grades: "
         << ptr->ShowSum() << endl;
    return 0;
}
```

```
The total grades: 190
destructor of 'Grade' works here
```

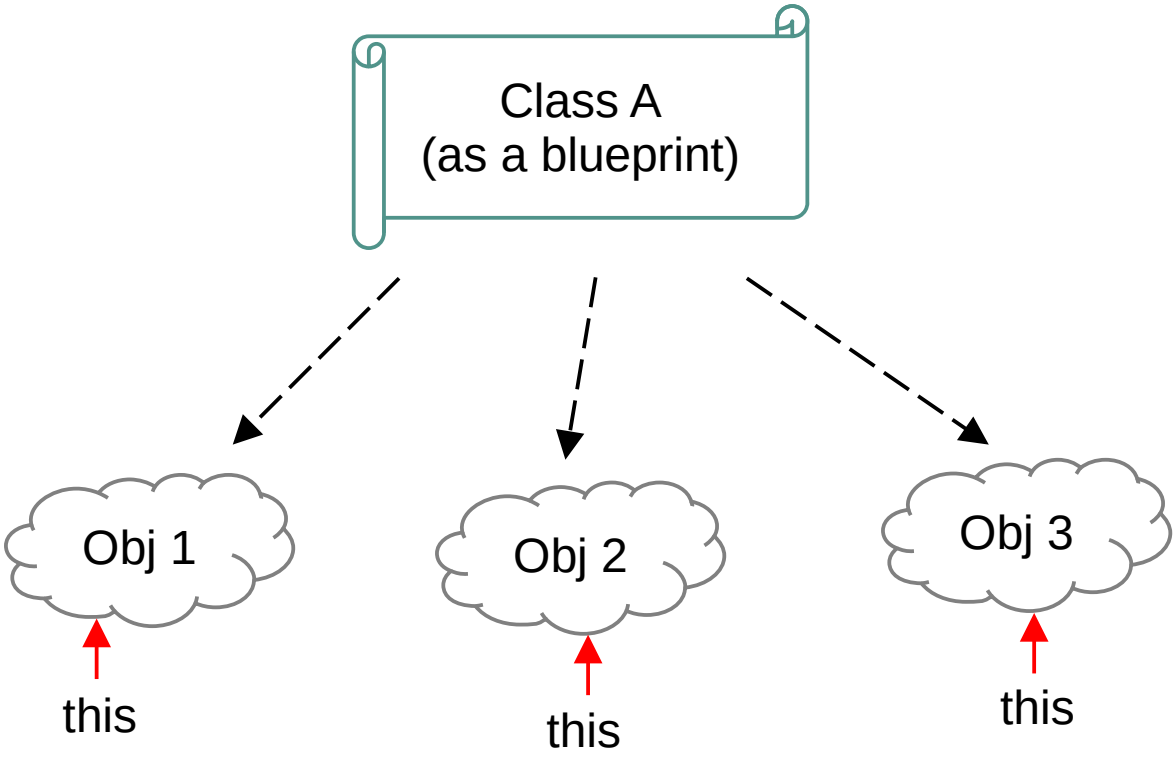
More Exercises

- https://onlinegdb.com/2oqsenisJp
- Design a **constructor** which can assign values of the data members of `Vehicle`.
- Prompt the user to input the number *n* of `vehicles`.
- **Use `new` and `delete` to construct a set of *n* vehicles.**
- Print all the vehicles with total prices and brands.

Sample input & output

```
2
Constructor works here!
100 200 300 Volswagen
Constructor works here!
200 300 400 BMW
VolswagenTotal price: 600
VolswagenTotal price: 900
destructor of 'Vehicle' works here
destructor of 'Vehicle' works here
```

Some notes on “this pointer”


Class A
(as a blueprint)
Obj 1
Obj 2
Obj 3
this
this
this

Example 1

```
class Demo {
private:
    int value;
public:
    Demo(int value) {
        this->value = value;
    // Using this pointer to refer to
    // the current object
    }
    void display() {
        cout << "Value: " << this->value << endl;
    }
};
```

```
int main() {
    Demo obj(10);
    obj.display(); // 10
    return 0;
}
```

Example 2

```
class Number {
private:
    int num;
public:
    Number(int num) {
        this->num = num;
    }

    Number& setValue(int num) {
        this->num = num;
        return *this; // Returning current object
    }

    void display() {
        cout << "Number: " << num << endl;
    }
};
```

```
int main() {
    Number obj(5);
    obj.setValue(10).display();
    return 0;
}
```

Example 3

```
class Employee {
private:
    string name;
    int age;
public:
    Employee(string name, int age) {
        this->name = name;
        this->age = age;
        // Resolving conflicts!
    }

    void show() {
        cout << "Employee Name: "
             << this->name << "("
             << this->age << ")" << endl;
    }
};
```

```
int main() {
    Employee emp("Alice", 30);
    emp.show(); // Alice(30)
    return 0;
}
```

Discussions & Questions